Problem 1: ℓ_1-Minimization Encourages Sparse

Claim: Let $A \in \mathbb{R}^{M\times N}$, $y \in \mathbb{R}^M$, and $x \in \mathbb{R}^N$. If there is a unique solution x_1 to the ℓ_1-minimization problem

$$(P_1): \quad \min_{x\in\mathbb{R}^N} \|x\|_1 \quad \text{subject to} \quad y = Ax,$$

then $\|x_1\|_0 \leq M$.

Proof. By contradiction. Suppose $\|x_1\|_0 > M$.

Since x_1 satisfies $Ax_1 = y$, we have a system of M linear equations in N unknowns.

If $\|x_1\|_0 > M$, then more than M components of x_1 are nonzero.

By the fundamental theorem of linear equations, with M equations and more than M active unknowns, the solution cannot be unique (there exist infinitely many solutions).

This contradicts the uniqueness of x_1.

Therefore, $\|x_1\|_0 \leq M$.

Problem 2: Cauchy-Schwarz Inequality

Claim: $|\langle u, v\rangle| \leq \|u\| \cdot \|v\|$

Proof. Case 1: If $v = 0$, then both sides equal zero.

Case 2: Assume $v \neq 0$. For any $t \in \mathbb{R}$:

$$\begin{aligned} 0 \leq \|u - tv\|^2 &= \langle u - tv, u - tv\rangle \\ &= \|u\|^2 - 2t\langle u, v\rangle + t^2\|v\|^2 \end{aligned}$$

This is a quadratic in t: $f(t) = \|v\|^2t^2 - 2\langle u, v\rangle t + \|u\|^2 \geq 0$ for all t.

For $f(t) \geq 0$ for all t, the discriminant must satisfy:

$$\begin{aligned} \Delta = 4\langle u, v\rangle^2 - 4\|v\|^2\|u\|^2 \leq 0 \\ \iff \langle u, v\rangle^2 \leq \|u\|^2\|v\|^2 \\ \iff |\langle u, v\rangle| \leq \|u\| \cdot \|v\| \end{aligned}$$

Problem 3: Reverse Triangle Inequality

Claim: $|\|u\| - \|v\|| \leq \|u - v\|$

Proof. By the standard triangle inequality:

$$\begin{aligned} \|u\| &= \|(u - v) + v\| \leq \|u - v\| + \|v\| \\ &\implies \|u\| - \|v\| \leq \|u - v\| \end{aligned} \tag{*}$$

Swapping u and v:

$$\|v\| - \|u\| \leq \|v - u\| = \|u - v\| \tag{**}$$

From $(*)$ and $(**)$:

$$\begin{aligned} -\|u - v\| &\leq \|u\| - \|v\| \leq \|u - v\| \\ &\iff |\|u\| - \|v\|| \leq \|u - v\| \end{aligned}$$

Problem 4

Claim: With the assumptions in Lemma 2.2, show that:

$$|\langle Au, Av \rangle| = \frac{1}{4} \left| \|Au + Av\|_2^2 - \|Au - Av\|_2^2 \right|$$

Proof. Expand $\|Au + Av\|_2^2$:

$$\begin{aligned} \|Au + Av\|_2^2 &= \langle Au + Av, Au + Av \rangle \\ &= \|Au\|_2^2 + 2\langle Au, Av \rangle + \|Av\|_2^2 \end{aligned}$$

Expand $\|Au - Av\|_2^2$:

$$\begin{aligned} \|Au - Av\|_2^2 &= \langle Au - Av, Au - Av \rangle \\ &= \|Au\|_2^2 - 2\langle Au, Av \rangle + \|Av\|_2^2 \end{aligned}$$

Compute the difference:

$$\begin{aligned} \|Au + Av\|_2^2 - \|Au - Av\|_2^2 &= 4\langle Au, Av \rangle \\ &\iff \langle Au, Av \rangle = \frac{1}{4} \left(\|Au + Av\|_2^2 - \|Au - Av\|_2^2 \right) \\ &\iff |\langle Au, Av \rangle| = \frac{1}{4} \left| \|Au + Av\|_2^2 - \|Au - Av\|_2^2 \right| \end{aligned}$$